CAMBRIDGE INTERNATIONAL EXAMINATIONS

GCE Ordinary Level

MARK SCHEME for the May/June 2014 series

5014 ENVIRONMENTAL MANAGEMENT

5014/21 Paper 2, maximum raw mark 60

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1 (a) (i) 85.4;17.08; (allow 85.5; 17.1) [2]

(ii) Ethiopia; highest population; so more fertile women; women have more children than other countries / ref to 5.97; AVP, e.g. a valid point about better health; [2]

(iii) difficult to get agreement between countries; environmental protests; who is going to invest / not enough revenue; IMF / world bank not supporting development; which country will be responsible for each section; lack of skills in countries (planning/building); ref. to war or civil war; AVP, such as political change; [3]

(iv) many people employed (to build the project); increased trade between countries; more trade within each country; so more jobs; government revenue; so better housing; electricity supply; more to spend on other infrastructure such as health; and schools; [3]

(v) more trade between countries; South Sudan / Ethiopia linked to sea port; so more exports possible; [2]

(vi) much faster; least amount of material moved; so cheaper; ref. to building in low biodiversity area; replanting / landscaping; pollution control; AVP; [3]

(b) (i) yes because the whole project needs a port; all three countries want to go ahead; only 400 ha are used so damage limited even though it is a world heritage site; too much time and money invested already to stop now; more employment;

OR no because large area of land destroyed; too expensive; damage trade in existing ports / eq.; AVP; [3]

(ii) to export for foreign exchange; to pay for the project; to use in local power stations in the future; reduce coal imports; [2]

2 (a) lung damage; long term poisoning / toxic gases; possible stunted growth; more lung infections / diseases; breathing difficulties / bronchitis / asthma; burns / skin infections; [3]

(b) (i) C A B D; *Accept A C B D.* [1]

(ii) 50 80(%);
40 and 50 (m);
Any two for one mark. [2]

(iii) heavy duty so more copper / more money / eq.; [1]

(c) (i) could pick up only light or heavy wire or one wire type / might not find much in five mins / not very representative of the dump / eq.; only one dump sampled; [1]

(ii) all types of wire in sample; more wire collected; more than one dump; more time; [2]

(iii) more dumps sampled (so more representative); each type of wire sorted / weighed separately; [2]

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(d) (i) *Two sensible questions.* Such as any deaths from cancer? How often has a member of the household been in hospital? Men / women died more? Age of death?
Ref. to other illness. [2]

(ii) Q with reason; [1]

(iii) blows from the east (to the west); so more air pollution at Q than the other settlements; [2]

(e) Q / R (Q and R); they are close to a water channel so more chance of contaminated water supply; or breeding mosquitoes; [2]

(f) (i) *advantages*: plastic does not need to be burn / improved air quality; plastic can be recycled; more copper recovered quickly;
disadvantages: cost of machines; maintenance of machines; need electrical power to work them; need for skilled labour; [3]

(ii) impossible to enforce; the waste has to dealt with; people have to make a living; more important things to tackle first; [2]

3 (a) (i) both axes labelled;; plots; key; [4]

(ii) plant height increases and then drops; so does the number of roots; ref. to figures; [3]

(iii) 4 (5) cowpeas per m^2 this gives the most roots to sell; above this and less roots and more top growth; [2]

(iv) cowpeas fix nitrogen in the soil / eq.; [1]

(v) the cost of fertiliser keeps going up; it is made using oil; so contributes to climate change; soil degraded by the same crop and fertiliser each year; AVP; [3]

(b) (i) 31.6 38.2; *Ignore units.* [1]

(ii) the yield with cowpeas is higher; in every year; the plots have different conditions but cowpeas still give higher yields; AVP; [2]

AVP = Alternative Valid Point.

[Total: 60]

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